

When Does OLS Break? Eigenvalue Spectra of Random Design Matrices and Their Impact on Linear Model Inference

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1 Introduction

Throughout the course [2], we have studied the linear model $\mathbb{E}(Y) = X\beta$ and derived foundational results: the Gauss–Markov theorem guarantees that the least squares estimator $\hat{\beta} = (X^\top X)^{-1}X^\top Y$ is the best linear unbiased estimator; confidence ellipsoids for β are governed by the eigenvalues of $X^\top X$; and diagnostic tools such as leverage and Cook’s distance depend on the hat matrix $P = X(X^\top X)^{-1}X^\top$.

All of these results rest on $X^\top X$ being well-conditioned. But what happens when the number of predictors p is not negligibly small compared to the number of observations n ? In modern applications; genomics, finance, text analysis, it is common for p/n to be substantial. The Marchenko–Pastur (MP) distribution [1] provides a precise description of this regime: when X has i.i.d. entries, the empirical spectral distribution of $\frac{1}{n}X^\top X$ converges to a known limiting law as $n, p \rightarrow \infty$ with $p/n \rightarrow \gamma$.

The question I investigate is: **how does the ratio $\gamma = p/n$ affect the reliability of OLS estimation, confidence regions, and diagnostic tools, and can the MP law predict this degradation?**

2 Why This Question

The linear model is arguably the most important tool in applied statistics, and much of this course has been devoted to understanding its properties. However, the course primarily treats the design matrix X as fixed and well-behaved; the results are derived for general X of full rank, without asking how those results perform for different kinds of X .

In practice, the design matrix is determined by the data we collect. When p is large relative to n (many parameters relative to number of datapoints), as is increasingly common in areas like finance (where portfolio optimisation involves many assets), the properties of $X^\top X$ become a genuine concern. As evidenced by my first coursework, my interests primarily lie in financial mathematics, so I was curious about whether there was a clean mathematical framework that

could quantify this, and came across the Marchenko–Pastur distribution which turns out to be exactly that.

The course also ends by posing the question of what to do when $p \gg n$ (Section 11.8). This project provides a partial answer by first explaining why things go wrong at a spectral level, before the practical fixes (ridge regression, LASSO) become necessary.

3 Background: The Marchenko–Pastur Law

Let $X \in \mathbb{R}^{n \times p}$ be a random matrix with i.i.d. entries satisfying $\mathbb{E}[X_{ij}] = 0$ and $\text{Var}(X_{ij}) = 1$. Define the sample covariance matrix $S_n = \frac{1}{n}X^\top X$. In the linear model with standardised predictors, $X^\top X = nS_n$ governs the covariance of $\hat{\beta}$, the shape of confidence ellipsoids, and the hat matrix.

Theorem 1 (Marchenko–Pastur [1]). *Suppose $n, p \rightarrow \infty$ with $p/n \rightarrow \gamma \in (0, \infty)$. The empirical spectral distribution of S_n converges almost surely to the distribution with density (for $\gamma \leq 1$)*

$$f_\gamma(\lambda) = \frac{1}{2\pi\gamma} \cdot \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda}, \quad \lambda \in [\lambda_-, \lambda_+],$$

where $\lambda_\pm = (1 \pm \sqrt{\gamma})^2$. When $\gamma > 1$, there is additionally a point mass of weight $1 - 1/\gamma$ at $\lambda = 0$.

The key features: as $\gamma \rightarrow 1$, $\lambda_- \rightarrow 0$, meaning the smallest eigenvalue approaches zero. For $\gamma > 1$, $X^\top X$ is singular and OLS is undefined. A remarkable property is universality: the limiting distribution depends only on γ , not on the entry distribution [3].

4 Connecting the MP Law to the Linear Model

4.1 Covariance of the Least Squares Estimator

From Section 9.7 of [2], the covariance of the least squares estimator under the second-order assumptions is $\text{cov}(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$. The total variance is

$$\text{trace}(\text{cov}(\hat{\beta})) = \frac{\sigma^2}{n} \sum_{i=1}^p \frac{1}{\mu_i},$$

where μ_i are the eigenvalues of S_n . By the MP law, this sum has a known limit [3]:

$$\frac{1}{p} \sum_{i=1}^p \frac{1}{\mu_i} \rightarrow \int_{\lambda_-}^{\lambda_+} \frac{1}{\lambda} f_\gamma(\lambda) d\lambda = \frac{1}{1 - \gamma}, \quad \text{for } \gamma < 1.$$

Therefore, the average variance per component of $\hat{\beta}$ scales as $\sigma^2/[n(1 - \gamma)]$, which diverges as $\gamma \rightarrow 1^-$.

4.2 Confidence Ellipsoids

From Section 10.5 of [2], the $1 - \alpha$ confidence region for β is an ellipsoid with half-axes proportional to $1/\sqrt{\lambda_i(X^\top X)}$. As $\gamma \rightarrow 1$, $\lambda_{\min} \approx n(1 - \sqrt{\gamma})^2 \rightarrow 0$, so the ellipsoid becomes extremely elongated, there are directions in parameter space along which the data provides almost no information.

4.3 Leverage and the Hat Matrix

From Section 11.2 of [2], the leverage of the i th observation is P_{ii} , the i th diagonal element of the hat matrix. Since $\text{trace}(P) = p$ (by Lemma 12 in [2]), the average leverage is $p/n = \gamma$. The course's rule of thumb flags observations with $P_{ii} > 2\gamma$. When γ is small, this is discriminating. When $\gamma \rightarrow 1$, the average leverage is already near 1, many observations exceed the threshold, and the diagnostic loses its power. In the extreme case $p = n$, every $P_{ii} = 1$ and the model interpolates the data perfectly.

5 Simulation Study

We verify the theoretical predictions via simulation. All code was implemented in Python using NumPy and Matplotlib and is available [here](#).

5.1 Eigenvalue Histograms vs. the MP Density

For each $\gamma \in \{0.2, 0.5, 0.8, 0.95\}$, we fix $n = 2000$, set $p = \lfloor \gamma n \rfloor$, generate X with i.i.d. $\mathcal{N}(0, 1)$ entries, and plot the eigenvalue histogram of S_n against the MP density. We also repeated this with Uniform and Rademacher entries and observed indistinguishable results.

Eigenvalue distribution of $\frac{1}{n}X^\top X$ vs. Marchenko-Pastur density

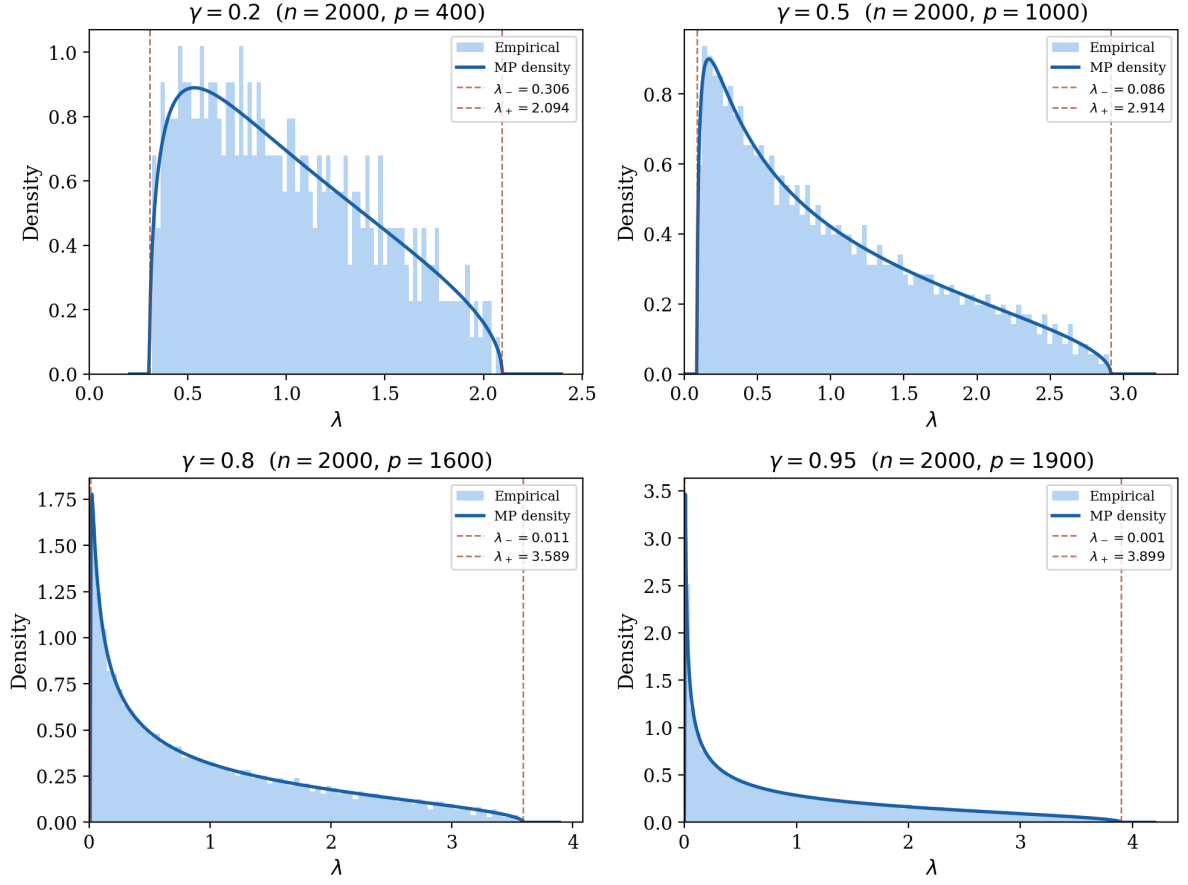


Figure 1: Eigenvalue histograms of $\frac{1}{n}X^\top X$ vs. the MP density for $n = 2000$. As γ increases, the support widens and λ_- (left dashed line) approaches zero.

5.2 Total Variance of $\hat{\beta}$

We simulate $Y = X\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I_n)$, $\beta = (1, 0, \dots, 0)^\top$, and $n = 500$. Over 200 replications per γ , we compute the average MSE $\frac{1}{p}\|\hat{\beta} - \beta\|^2$ and compare it to the MP prediction $\sigma^2/[n(1 - \gamma)]$.

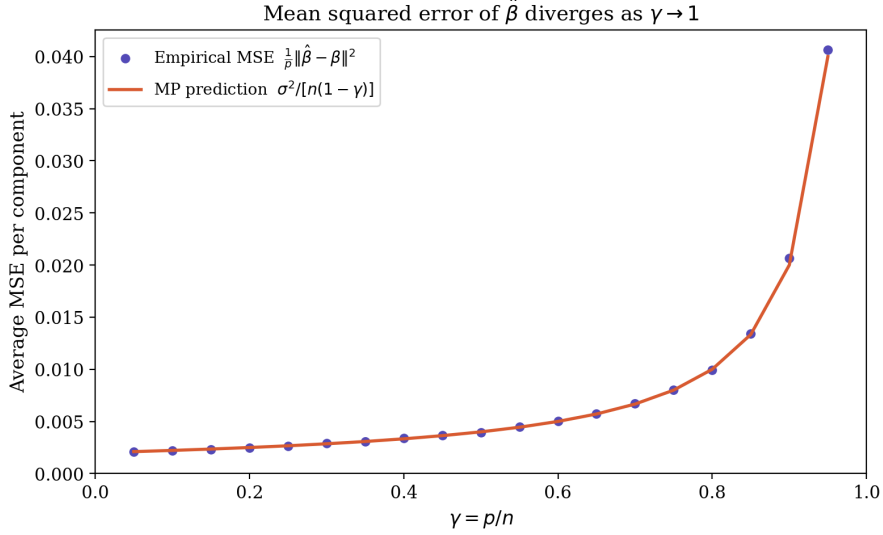


Figure 2: Empirical MSE of $\hat{\beta}$ vs. the MP prediction $\sigma^2/[n(1-\gamma)]$. The near-perfect agreement confirms that the MP law accurately describes variance inflation.

5.3 Confidence Ellipsoid Shape

We examine the condition number $\lambda_{\max}/\lambda_{\min}$ of $X^\top X$ and the longest half-axis of the 95% confidence ellipsoid (proportional to $1/\sqrt{\lambda_{\min}}$). The MP law predicts the condition number as $(1 + \sqrt{\gamma})^2/(1 - \sqrt{\gamma})^2$.

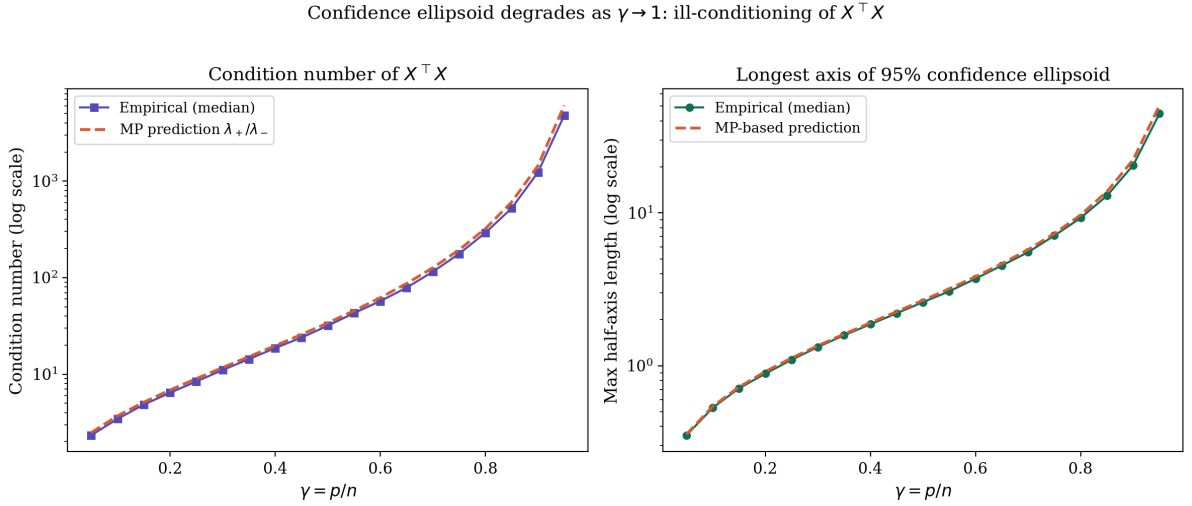


Figure 3: Left: condition number of $X^\top X$ matches the MP prediction. Right: the longest half-axis of the 95% confidence ellipsoid diverges correspondingly. Both use $n = 500$, median over 100 replications.

5.4 Leverage Distribution

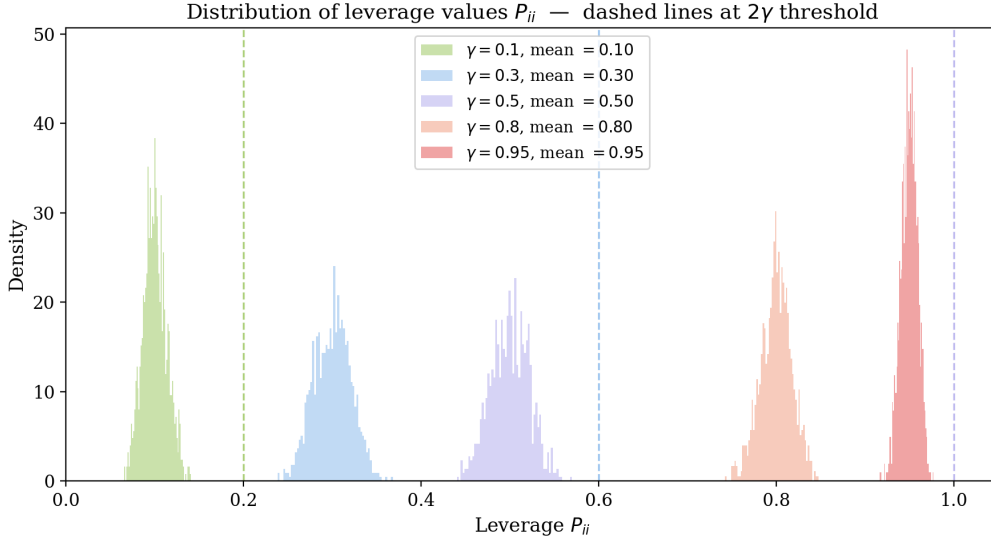


Figure 4: Distribution of leverage values P_{ii} for $n = 1000$. Dashed lines mark the 2γ diagnostic threshold from [2]. At high γ , the spread overwhelms the threshold.

At small γ , leverages concentrate tightly around γ . As γ grows, the distribution spreads out, and the 2γ threshold flags an increasing fraction of observations, rendering it less useful.

5.5 Confidence Interval Coverage and Width

Under normal theory assumptions, the t -based CI for β_j has exact 95% coverage for all γ (via the exact pivot $(\hat{\beta}_j - \beta_j)/\hat{SE} \sim t_{n-p}$). However, the MP law predicts the CI width diverges as $1/\sqrt{1-\gamma}$. The CI becomes valid but useless.

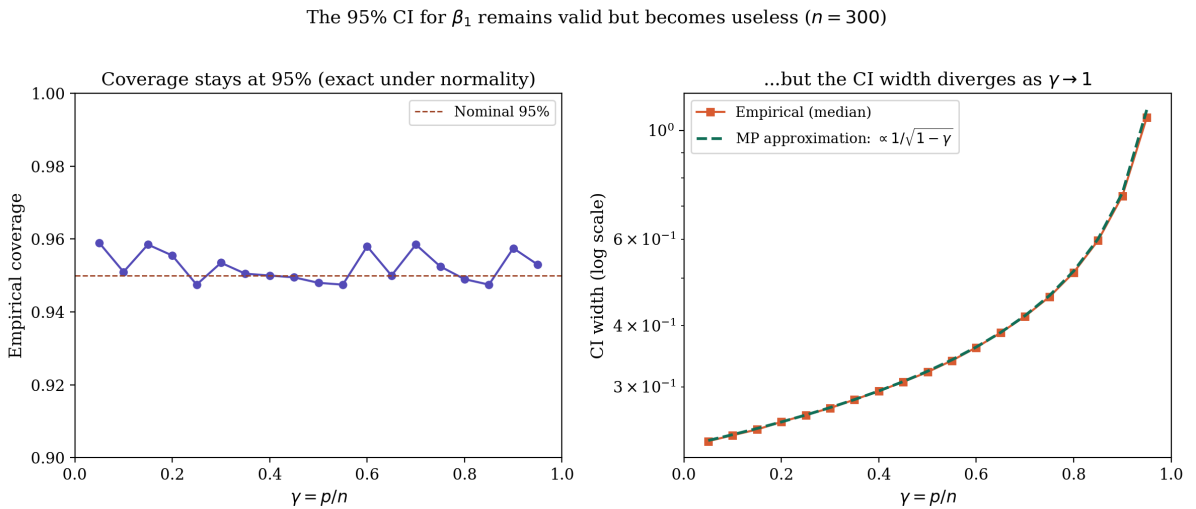


Figure 5: Left: 95% CI coverage remains nominal for all γ . Right: CI width diverges as $\gamma \rightarrow 1$, matching the MP approximation. ($n = 300$, 2000 replications.)

6 The $\gamma \geq 1$ Cliff

When $p > n$, $X^\top X$ has rank at most n , so at least $p - n$ eigenvalues are exactly zero and OLS is undefined. The MP law predicts this precisely: a point mass of weight $1 - 1/\gamma$ at zero. This is the setting raised in Section 11.8 of [2]. The MP law explains why things fail, motivating ridge regression (which shifts eigenvalues away from zero) and the LASSO (which enforces sparsity).

The $\gamma \geq 1$ cliff: $X^\top X$ is singular, OLS is undefined

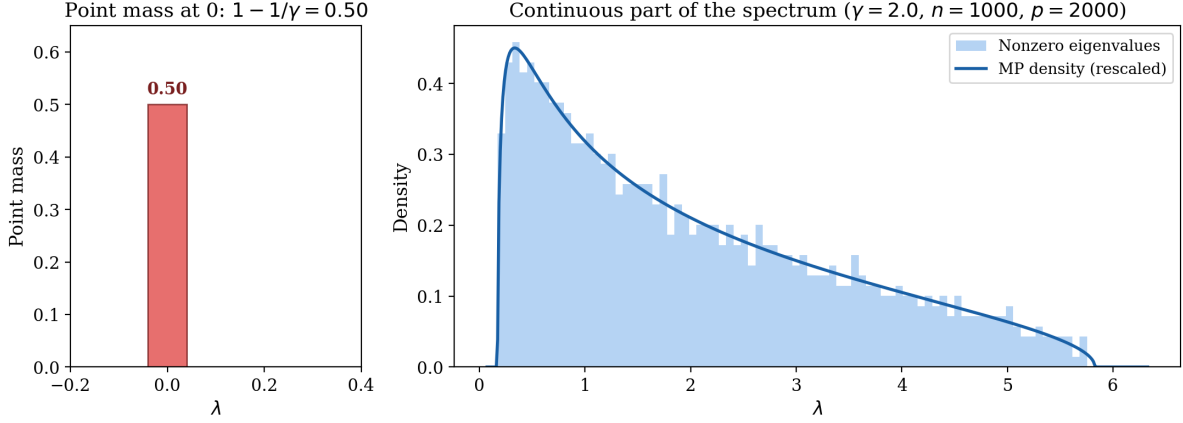


Figure 6: At $\gamma = 2$: half the eigenvalues are exactly zero (left), while the nonzero eigenvalues follow the continuous MP density (right). OLS is undefined.

7 Critical Use of AI Tools

AI was again used to make connections between the course and the MP law, which I had read about in my own time, and provide a reasonable topic for the coursework. I also was assisted by AI in the link between limit and integral in section 4.1, but verified independently too. Additionally, some coding aspects were aided by AI.

8 Conclusion

We have explored how $\gamma = p/n$ affects OLS in the linear model. The MP law [1] describes the eigenvalue spectrum of $\frac{1}{n}X^\top X$ and governs: the total variance of $\hat{\beta}$, which scales as $\sigma^2/[n(1-\gamma)]$ and diverges as $\gamma \rightarrow 1$; the condition number and shape of confidence ellipsoids; the width of individual CIs (which diverge even though coverage remains exact); the distribution of leverage values; and the complete breakdown at $\gamma \geq 1$. Simulations confirm all predictions, and universality was verified for non-Gaussian entries.

8.1 Limitations and Future Work

The MP law applies to random X with i.i.d. entries. Real design matrices often have correlated columns; generalised MP results exist for these settings [3] but are beyond the scope of the

coursework. A natural extension would be to apply these ideas to a real high-dimensional dataset, or to study how ridge regularisation shifts the MP spectrum to restore well-conditioned inference.

References

- [1] V. A. Marchenko and L. A. Pastur, “Distribution of eigenvalues for some sets of random matrices,” *Mat. Sb.*, vol. 72, no. 4, pp. 507–536, 1967.
- [2] R. Passeggeri, *Statistical Modelling I (MATH 50011) Lecture Notes*, Imperial College London, Spring 2026.
- [3] Z. D. Bai and J. W. Silverstein, *Spectral Analysis of Large Dimensional Random Matrices*, 2nd ed., Springer, 2010.